

LEDGER · 2026-08-29 · BIOMEDICAL MATERIALS

Sustained-Release Silk Fibroin Microneedle Matrices via Lithium Bromide Extraction

PRODUCT



A product dossier of The Absence Audit. The current audit flags this concept — Margin fails the operating-cost check — and the full disclosure ships inside the dossier.

Incumbent benchmark

Hyaluronic Acid (HA) microneedle matrices

Entry application

Transdermal sustained-release hormonal contraception and prophylactic antiretrovirals

Measured advantage	>100x increase in delivery duration (up to 100 days vs. <60 minutes for HA)
Time to first revenue	60 months
Gross margin at parity	43.3%
Startup CapEx	\$15,400
Payback from first sale	52.7 months
Capital productivity	4.09x
Regulatory risk	High — regulatory qualification is a material barrier (\$5,000,000 estimated)

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Mechanism, operating protocol with real conditions and yields, computed unit economics with a six-scenario stress test, the absence audit, and every resolved citation.

[Buy the full dossier — \\$499.00 →](#)

Assessed 2026-08-29 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

Terms

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